

Lecture No. 12

Fundamentals of Probability

PART-1

PROBABILITY

Probability is the likelihood that an event will occur under a set of given conditions. The probability of an event occurring has a value between 0 and 1. An “impossible event” would have a probability of 0; a “certain event” would have a probability of 1.

$$0 \leq P(A) \leq 1.$$

Classical (Theoretical) Probability Formula

For the Classical Probability Formula, the outcomes must be equally likely. If the outcomes are not equally likely, then the Empirical Probability Formula should be used.

$$P(A) = \frac{s}{n} = \frac{\text{(Number of ways event A can occur)}}{\text{(Total number of all outcomes)}}$$

Example: What is the probability of drawing a 7 from a standard deck of 52 cards?

Solution: In a standard deck of cards, there are suits of cards each: Spades (black), Clubs (black), Hearts (red), and Diamonds (red). Each suit contains the following: the numbers through , Jack, Queen, King, and Ace. Therefore, a standard deck of cards contains four 7s.

$$P(\text{drawing a } 7) = \frac{4}{52} = \frac{1}{13} = .0769$$

Empirical Probability Formula (Actual data)

$$P(A) = \frac{(\text{Number of times event } A \text{ occurred})}{(\text{Number of times experiment performed})}$$

A die is rolled once. Find the probability of getting a 5.

Solution : There are six possible ways in which a die can fall, out of these only one is favourable to the event.

$$\therefore P(5) = \frac{1}{6}.$$

A coin is tossed once. What is the probability of the coin coming up with head ?

Solution : The coin can come up either 'head' (H) or a tail (T). Thus, the total possible outcomes are two and one is favourable to the event.

$$\text{So, } P(H) = \frac{1}{2}$$

A die is rolled once. What is the probability of getting a prime number ?

Solution : There are six possible outcomes in a single throw of a die. Out of these; 2, 3 and 5 are the favourable cases.

$$\therefore P(\text{Prime Number}) = \frac{3}{6} = \frac{1}{2}$$

A die is rolled once. What is the probability of the number '7' coming up ?

Solution : There are six possible outcomes in a single throw of a die and there is no face of the die with mark 7.

$$\therefore P(\text{number } 7) = \frac{0}{6} = 0$$

[Note : That the probability of impossible event is zero]

**In a simultaneous toss of two coins, find the probability of
(i) getting 2 heads (ii) exactly 1 head**

Solution : Here, the possible outcomes are
HH, HT, TH, TT.

i.e., Total number of possible outcomes = 4.

(i) Number of outcomes favourable to the event (2 heads) = 1 (i.e., HH).

$$\therefore P(2 \text{ heads}) = \frac{1}{4}.$$

(ii) Now the event consisting of exactly one head has two favourable cases, namely HT and TH.

$$\therefore P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}.$$

In a single throw of two dice, what is the probability that the sum is 9?

Solution : The number of possible outcomes is $6 \times 6 = 36$. We write them as given below :

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

Now, how do we get a total of 9. We have :

$$3 + 6 = 9$$

$$4 + 5 = 9$$

$$5 + 4 = 9$$

$$6 + 3 = 9$$

In other words, the outcomes (3, 6), (4, 5), (5, 4) and (6, 3) are favourable to the said event, i.e., the number of favourable outcomes is 4.

$$\text{Hence, } P(\text{a total of } 9) = \frac{4}{36} = \frac{1}{9}$$

From a bag containing 10 red, 4 blue and 6 black balls, a ball is drawn at random.

What is the probability of drawing

(i) a red ball ?

(ii) a blue ball ?

(iii) not a black ball ?

Solution : There are 20 balls in all. So, the total number of possible outcomes is 20. (Random drawing of balls ensure equally likely outcomes)

(i) Number of red balls = 10

$$\therefore P(\text{a red ball}) = \frac{10}{20} = \frac{1}{2}$$

(ii) Number of blue balls = 4

$$\therefore P(\text{a blue ball}) = \frac{4}{20} = \frac{1}{5}$$

(iii) Number of balls which are not black = $10 + 4$
 $= 14$

$$\therefore P(\text{not a black ball}) = \frac{14}{20} = \frac{7}{10}$$

A card is drawn at random from a well shuffled deck of 52 cards.

If A is the event of getting a queen and B is the event of getting a card bearing a number greater than 4 but less than 10, find $P(A)$ and $P(B)$.

Solution : Well shuffled pack of cards ensures equally likely outcomes.

∴ the total number of possible outcomes is 52.

(i) There are 4 queens in a pack of cards.

$$\therefore P(A) = \frac{4}{52} = \frac{1}{13}$$

(ii) The cards bearing a number greater than 4 but less than 10 are 5,6, 7,8 and 9.

Each card bearing any of the above number is of 4 suits diamond, spade, club or heart.

Thus, the number of favourable outcomes = $5 \times 4 = 20$

$$\therefore = \frac{20}{52} = \frac{5}{13}$$

Also Check.....

1. A die is rolled once. Find the probability of getting 3.
2. A coin is tossed once. What is the probability of getting the tail ?
3. What is the probability of the die coming up with a number greater than 3 ?
4. In a simultaneous toss of two coins, find the probability of getting 'at least' one tail.
5. From a bag containing 15 red and 10 blue balls, a ball is drawn 'at random'. What is the probability of drawing (i) a red ball ? (ii) a blue ball ?
6. If two dice are thrown, what is the probability that the sum is (i) 6 ? (ii) 8? (iii) 10? (iv) 12?
7. If two dice are thrown, what is the probability that the sum of the numbers on the two faces is divisible by 3 or by 4 ?
8. If two dice are thrown, what is the probability that the sum of the numbers on the two faces is greater than 10 ?
9. What is the probability of getting a red card from a well shuffled deck of 52 cards ?
10. If a card is selected from a well shuffled deck of 52 cards, what is the probability of drawing
 - (i) a spade ? (ii) a king ? (iii) a king of spade ?
11. A pair of dice are thrown. Find the probability of getting
 - (i) a sum as a prime number
 - (ii) a doublet, i.e., the same number on both dice
 - (iii) a multiple of 2 on one die and a multiple of 3 on the other.
12. Three coins are tossed simultaneously. Find the probability of getting
 - (i) no head (ii) at least one head (iii) all heads

CALCULATION OF PROBABILITY USING COMBINATORICS (PERMUTATIONS AND COMBINATIONS)

IN THIS SECTION :-

- we calculated the probability of an event by listing down all the possible outcomes and the outcomes favorable to the event.
- This is possible when the number of outcomes is small, otherwise it becomes difficult and time consuming process.
- In general, we do not require the actual listing of the outcomes, but require only the total number of possible outcomes and the number of outcomes favorable to the event.
- In many cases, these can be found by applying the knowledge of permutations and combinations, which you have already studied.

A bag contains 3 red, 6 white and 7 blue balls. What is the probability that two balls drawn are white and blue ?

Solution : Total number of balls = $3 + 6 + 7 = 16$

Now, out of 16 balls, 2 can be drawn in ${}^{16}C_2$ ways.

$$\therefore \text{Exhaustive number of cases} = {}^{16}C_2 = \frac{16 \times 15}{2} = 120$$

Out of 6 white balls, 1 ball can be drawn in 6C_1 ways and out of 7 blue balls, one can be drawn in 7C_1 ways. Since each of the former case is associated with each of the later case, therefore total number of favourable cases are ${}^6C_1 \times {}^7C_1 = 6 \times 7 = 42$.

$$\therefore \text{Required probability} = \frac{42}{120} = \frac{7}{20}$$

Remarks

When two or more balls are drawn from a bag containing several balls, there are two ways in which these balls can be drawn.

- (i) **Without replacement** : The ball first drawn is not put back in the bag, when the second ball is drawn. The third ball is also drawn without putting back the balls drawn earlier and so on. Obviously, the case of drawing the balls without replacement is the same as drawing them together.
- (ii) **With replacement** : In this case, the ball drawn is put back in the bag before drawing the next ball. Here the number of balls in the bag remains the same, every time a ball is drawn.

In these types of problems, unless stated otherwise, we consider the problem of without replacement.

Find the probability of getting both red balls, when from a bag containing 5 red and 4 black balls, two balls are drawn,

(i) with replacement.

(ii) without replacement.

Solution : (i) Total number of balls in the bag in both the draws $= 5 + 4 = 9$

Hence, by fundamental principle of counting, the total number of possible outcomes $= 9 \times 9 = 81$.

Similarly, the number of favourable cases $= 5 \times 5 = 25$.

Hence, probability (both red balls) $= \frac{25}{81}$.

(ii) Total number of possible outcomes is equal to the number of ways of selecting 2 balls out of 9 balls $= {}^9C_2$.

Number of favourable cases is equal to the number of ways of selecting

2 balls out of 5 red balls $= {}^5C_2$.

$$\text{Hence, } P(\text{both red balls}) = \frac{{}^5C_2}{{}^9C_2} = \frac{\frac{5 \times 4}{1 \times 2}}{\frac{9 \times 8}{1 \times 2}} = \frac{5}{18}$$

Six cards are drawn at random from a pack of 52 cards. What is the probability that 3 will be red and 3 black?

Solution : Six cards can be drawn from the pack of 52 cards in ${}^{52}C_6$ ways.

i.e., Total number of possible outcomes = ${}^{52}C_6$

3 red cards can be drawn in ${}^{26}C_3$ ways and

3 black cards can be drawn in ${}^{26}C_3$ ways.

$\therefore$ Total number of favourable cases = ${}^{26}C_3 \times {}^{26}C_3$

Hence, the required probability = $\frac{{}^{26}C_3 \times {}^{26}C_3}{{}^{52}C_6} = \frac{13000}{39151}$

Four persons are chosen at random from a group of 3 men, 2 women and 4 children. Show that the chance that exactly two of them will be children is $\frac{10}{21}$.

Solution : Total number of persons in the group = $3 + 2 + 4 = 9$. Four persons are chosen at random. If two of the chosen persons are children, then the remaining two can be chosen from 5 persons (3 men + 2 women).

Number of ways in which 2 children can be selected from 4

$$\text{children} = {}^4C_2 = \frac{4 \times 3}{1 \times 2} = 6$$

Number of ways in which remaining of the two persons can be selected

$$\text{from 5 persons} = {}^5C_2 = \frac{5 \times 4}{1 \times 2} = 10$$

Total number of ways in which 4 persons can be selected out of

$$9 \text{ persons} = {}^9C_4 = \frac{9 \times 8 \times 7 \times 6}{1 \times 2 \times 3 \times 4} = 126$$

$$\text{Hence, the required probability} = \frac{{}^4C_2 \times {}^5C_2}{{}^9C_4} = \frac{6 \times 10}{126} = \frac{10}{21}$$

Three cards are drawn from a well-shuffled pack of 52 cards. Find the probability that they are a king, a queen and a jack.

Solution : From a pack of 52 cards, 3 cards can be drawn in ${}^{52}C_3$ ways, all being equally likely.

$\therefore$ Exhaustive number of cases = ${}^{52}C_3$

A pack of cards contains 4 kings, 4 queens and 4 jacks. A king, a queen and a Jack can each be drawn in 4C_1 ways and since each way of drawing a king can be associated with each of the ways of drawing a queen and a jack, the total number of favourable cases = ${}^4C_1 \times {}^4C_1 \times {}^4C_1$

$$\begin{aligned}\therefore \text{ Required probability} &= \frac{{}^4C_1 \times {}^4C_1 \times {}^4C_1}{{}^{52}C_3} \\ &= \frac{4 \times 4 \times 4}{52 \times 51 \times 50} \\ &\quad \frac{1 \times 2 \times 3}{1 \times 2 \times 3} \\ &= \frac{16}{5525}\end{aligned}$$

ADDITIVE LAW OF PROBABILITY

Let A be the event of getting an odd number and B be the event of getting a prime number in a single throw of a die. What will be the probability that it is either an odd number or a prime number ?

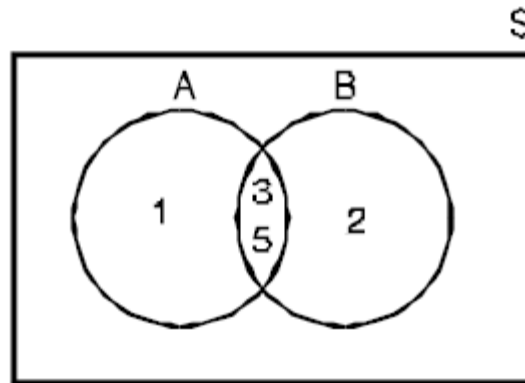
In a single throw of a die, the sample space would be

$$S = \{ 1, 2, 3, 4, 5, 6 \}$$

The outcomes favourable to the events A and B are

$$A = \{ 1, 3, 5 \}$$

$$B = \{ 2, 3, 5 \}$$



The outcomes favourable to the event 'A or B' are

$$A \cup B = \{1, 2, 3, 5\}.$$

Thus, the probability of getting either an odd number or a prime number will be

$$P(A \text{ or } B) = \frac{4}{6} = \frac{2}{3}$$

To Discover an alternative way.....

To discover an alternate method, we can proceed as follows :

The outcomes favourable to the event A are 1, 3 and 5.

$$\therefore P(A) = \frac{3}{6}$$

Similarly,
$$P(B) = \frac{3}{6}$$

The outcomes favourable to the event 'A and B' are 3 and 5.

$$\therefore P(A \text{ and } B) = \frac{2}{6}$$

$$\text{Now, } P(A) + P(B) - P(A \text{ and } B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6}$$

$$\frac{4}{6} = \frac{2}{3}$$

$$= P(A \text{ or } B)$$

Therefore

For any two events A and B of a sample space S,

$$P (A \text{ or } B) = P (A) + P (B) - P (A \text{ and } B)$$

or

$$P (A \cup B) = P (A) + P (B) - P (A \cap B)$$

A card is drawn from a well-shuffled deck of 52 cards. What is the probability that it is either a spade or a king ?

Solution : If a card is drawn at random from a well-shuffled deck of cards, the likelihood of any of the 52 cards being drawn is the same. Obviously, the sample space consists of 52 sample points.

If A and B denote the events of drawing a 'spade card' and a 'king' respectively, then the event A consists of 13 sample points, whereas the event B consists of 4 sample points. Therefore,

$$P(A) = \frac{13}{52}, \quad P(B) = \frac{4}{52}$$

The compound event $(A \cap B)$ consists of only one sample point, viz.; king of spade. So,

$$P(A \cap B) = \frac{1}{52}$$

Hence, the probability that the card drawn is either a spade or a king is given by

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} \\ &= \frac{16}{52} = \frac{4}{13} \end{aligned}$$

In an experiment with throwing 2 fair dice, consider the events

A : The sum of numbers on the faces is 8

B : Doubles are thrown.

What is the probability of getting A or B ?

Solution : In a throw of two dice, the sample space consists of $6 \times 6 = 36$ sample points.

The favourable outcomes to the event A (the sum of the numbers on the faces is 8) are

$$A = \{ (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) \}$$

The favourable outcomes to the event B (Double means both dice have the same number) are

$$B = \{ (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) \}$$

$$A \cap B = \{ (4,4) \}.$$

$$\text{Now } P(A) = \frac{5}{36}, \quad P(B) = \frac{6}{36}, \quad P(A \cap B) = \frac{1}{36}$$

Thus, the probability of A or B is

$$\begin{aligned} P(A \cup B) &= \frac{5}{36} + \frac{6}{36} - \frac{1}{36} \\ &= \frac{10}{36} = \frac{5}{18} \end{aligned}$$

ADDITIVE LAW OF PROBABILITY FOR MUTUALLY EXCLUSIVE EVENTS

We know that the events A and B are mutually exclusive, if and only if they have no outcomes in common. That is, for mutually exclusive events,

$$P (A \text{ and } B) = 0$$

Substituting this value in the additive law of probability, we get the following law :

$$P (A \text{ or } B) = P (A) + P (B)$$

In a single throw of two dice, find the probability of a total of 9 or 11.

Solution : Clearly, the events - a total of 9 and a total of 11 are mutually exclusive.

Now
$$P(\text{a total of 9}) = P[(3, 6), (4, 5), (5, 4), (6, 3)] = \frac{4}{36}$$

$$P(\text{a total of 11}) = P[(5, 6), (6, 5)] = \frac{2}{36}$$

Thus,
$$P(\text{a total of 9 or 11}) = \frac{4}{36} + \frac{2}{36}$$
$$= \frac{1}{6}$$

The probabilities that a student will receive an A, B , C or D grade are 0.30, 0.35, 0.20 and 0.15 respectively. What is the probability that a student will receive at least a B grade ?

Solution : The event at least a 'B' grade means that the student gets either a B grade or an A grade.

$$\begin{aligned}\therefore P(\text{at least B grade}) &= P(\text{B grade}) + P(\text{A grade}) \\ &= 0.35 + 0.30 \\ &= 0.65\end{aligned}$$

MULTIPLICATION LAW OF PROBABILITY FOR INDEPENDENT EVENTS

Let us recall the definition of independent events.

Two events A and B are said to be independent, if the occurrence or nonoccurrence of one does not affect the probability of the occurrence (and hence non-occurrence) of the other.

Can you think of some examples of independent events ?

- The event of getting 'H' on first coin and the event of getting 'T' on the second coin in a simultaneous toss of two coins are independent events.
- What about the event of getting 'H' on the first toss and event of getting 'T' on the second toss in two successive tosses of a coin ? They are also independent events.

Note : If the cards are drawn with replacement, then the two events become independent. Is there any rule by which we can say that the events are independent ?
How to find the probability of simultaneous occurrence of two independent events?
If A and B are independent events, then

$$P (A \text{ and } B) = P(A) \cdot P (B)$$

or

$$P(A \cap B) = P(A) \cdot P (B)$$

Thus, the probability of simultaneous occurrence of two independent events is the product of their separate probabilities.

Note : The above law can be extended to more than two independent events, i.e.,

$$P(A \cap B \cap C...) = P(A) \cdot P(B) \cdot P(C)...$$

A die is tossed twice. Find the probability of a number greater than 4 on each throw.

Solution : Let us denote by A, the event 'a number greater than 4' on first throw. B be the event 'a number greater than 4' in the second throw. Clearly A and B are independent events.

In the first throw, there are two outcomes, namely, 5 and 6 favourable to the event A.

$$\therefore P(A) = \frac{2}{6} = \frac{1}{3}$$

Similarly,

$$P(B) = \frac{1}{3}$$

Hence, $P(A \text{ and } B) = P(A) \cdot P(B)$

$$= \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}.$$

Arun and Tarun appear for an interview for two vacancies. The probability of Arun's selection is $\frac{1}{3}$ and that of Tarun's selection is $\frac{1}{5}$.

Find the probability that:

- (a) both of them will be selected.**
- (b) none of them is selected.**
- (c) at least one of them is selected.**
- (d) only one of them is selected.**

Solution : Probability of Arun's selection = $P(A) = \frac{1}{3}$
 Probability of Tarun's selection = $P(T) = \frac{1}{5}$

(a) $P(\text{both of them will be selected}) = P(A) P(T)$

$$= \frac{1}{3} \times \frac{1}{5}$$

$$= \frac{1}{15}$$

(b) $P(\text{none of them is selected})$

$$= P(\bar{A}) P(\bar{T})$$

$$= \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right)$$

$$= \frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

(c) $P(\text{at least one of them is selected})$
 $= 1 - P(\text{None of them is selected})$

$$= 1 - P(\bar{A}) P(\bar{T})$$

$$= 1 - \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right)$$

$$= 1 - \left(\frac{2}{3} \times \frac{4}{5}\right)$$

$$= 1 - \frac{8}{15} = \frac{7}{15}$$

(d) $P(\text{only one of them is selected})$

$$= P(A) P(\bar{T}) + P(\bar{A}) P(T)$$

$$= \frac{1}{3} \times \frac{4}{5} + \frac{2}{3} \times \frac{1}{5}$$

$$= \frac{6}{15} = \frac{2}{5}$$

PART-2

CONDITIONAL PROBABILITY

The sample space for an experiment must often be changed when some additional information pertaining to the outcome of the experiment is received.

The effect of such information is to REDUCE the sample space by excluding some outcomes as being impossible which BEFORE receiving the information were believed possible.

The probabilities associated with such a reduced sample space are called conditional probabilities.

The following example illustrates the concept of conditional probability:

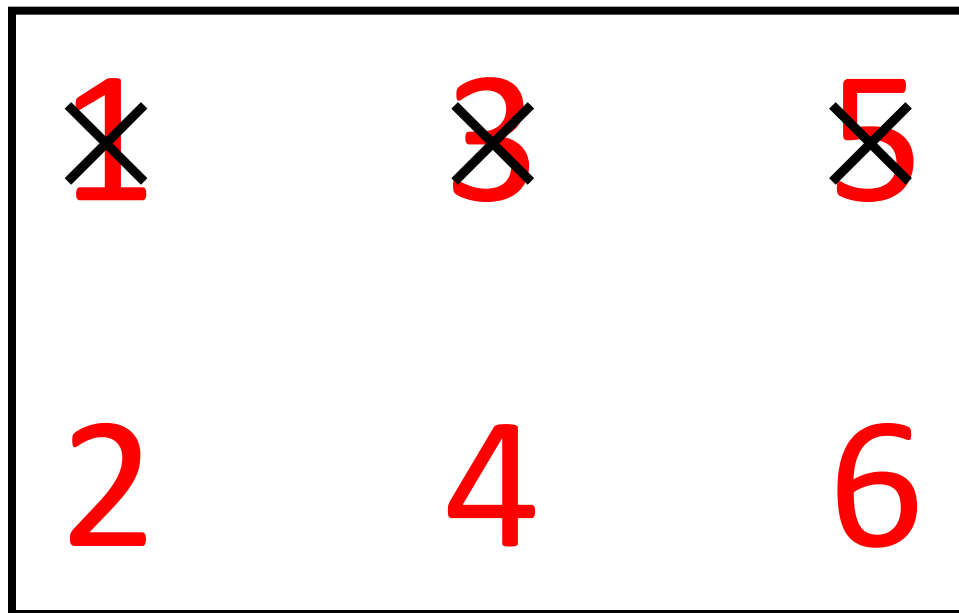
EXAMPLE

Suppose that we toss a fair die. Then the sample space of this experiment is $S = \{1,2,3,4,5,6\}$.

Suppose we wish to know the probability of the outcome that the die shows 6 (say event A).

Also, suppose that, before seeing the outcome, we are told that the die shows an EVEN number of dots (say event B).

Then the information that the die shows an even number excludes the outcomes 1, 3 and 5, and thereby reduces the original sample space to a sample space that consists of three outcomes 2, 4 and 6, i.e. the reduced sample space is $B = \{2, 4, 6\}$.



(The sample space is reduced.)

Then, the desired probability in the reduced sample space B is $1/3$ (since each outcome in the reduced sample space is EQUALLY LIKELY).

This probability $1/3$ is called the conditional probability of the event A because it is computed under the CONDITION that the die has shown an even number of dots.

In other words,

$$P(\text{die shows 6} / \text{die shows even numbers}) = 1/3,$$

Sometimes, it is not very convenient to compute a conditional probability by first determining the number of sample points that belong to the reduced sample space.

In such a situation, we can utilize the following alternative method of computing a conditional probability:

CONDITIONAL PROBABILITY

If A and B are two events in a sample space S and if P(B) is not equal to zero, then the conditional probability of the event A given that event B has occurred, written as P(A/B), is defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

where $P(B) > 0$.

(If $P(B) = 0$, the conditional probability P(A/B) remains undefined.)

Similarly

$$P(B / A) = \frac{P(A \cap B)}{P(A)}$$

where $P(A) > 0$.

It should be noted that $P(A/B)$ SATISFIES all the basic axioms of probability, namely:

- i) $0 \leq P(A/B) \leq 1.$
- ii) $P(S / B) = 1$
- iii) $P(A_1 \cup A_2 / B) = P(A_1 / B) + P(A_2 / B)$

(provided that the events A_1 and A_2 are mutually exclusive).

Example

A man tosses two fair dice. What is the conditional probability that the sum of the two dice will be 7 given that

- (i) The sum is greater than 6
- (ii) The two dice had the same outcome

Solution

The sample space S for this experiment consists of the following 36 equally likely outcomes:

$$S = \left\{ \begin{array}{cccccc} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (6,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

Let $A = \{\text{sum is } 7\}$

Then

$$A = \{(1,6) \ (2,5) \ (3,4) \ (4,3) \ (5,2) \ (6,1)\}$$

The total no. ordered pairs in this set is 6.

Hence, we have:

$$P(A) = 6/36$$

$$\text{Let } B = \{\text{sum greater than } 6\}$$

$$B = \left\{ \begin{array}{cccccc} & & & & & (1,6) \\ & & & & (2,5) & (2,6) \\ & & (3,4) & (6,5) & (3,6) & \\ & (4,3) & (4,4) & (4,5) & (4,6) & \\ (5,2) & (5,3) & (5,4) & (5,5) & (5,6) & \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{array} \right\}$$

Therefore

The total no. ordered pairs in this set is 21.

Hence, we have:

$$P(B) = 21/36$$

Let

$C = \{\text{the two dice have the same outcome}\}$

Therefore,

$$C = \{(1,1) \ (2,2) \ (3,3) \ (4,4) \ (5,5) \ (6,6)\}$$

The total no. ordered pairs in this set is 6.

Hence, we have:

$$P(C) = 6/36$$

The conditional probabilities $P(A/B)$ and $P(A/C)$ are given by the formulae:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A/C) = \frac{P(A \cap C)}{P(C)}$$

In order to compute $P(A/B)$, we first need to find

$P(A \cap B)$, and, for this purpose, we first need to determine the number of ordered pairs that belong to

this event:

$$A \cap B = \{(1,6) \ (2,5) \ (3,4) \ (4,3) \ (5,2) \ (6,1)\}$$

The total no. ordered pairs in this set is 6.

Hence, we have:

$$P(A \cap B) = 6 / 36$$

Hence, using the definition of conditional probability, we get

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{6}{36}}{\frac{21}{36}} = \frac{2}{7}$$

In order to compute $P(A/C)$, we first need to find $P(A \cap C)$, and for this purpose, we first need to determine the number of ordered pairs that belong to this event:

$$A \cap C = \phi$$

This is a null set because, obviously, none of the six ordered pairs is such whose sum is 7.

$$(1,1) \quad (2,2) \quad (3,3) \quad (4,4) \quad (5,5) \quad (6,6)$$

Since $A \cap C$ is a null set (an empty set), hence the total number of ordered pair in this set is 0.

Hence,

$$P(A \cap C) = 0 / 36 = 0$$

$$\text{Conditional Probability} = \frac{\text{Probability of joint event}}{\text{Probability of conditioning event}}$$

i) $0 \leq P(A/B) \leq 1.$

ii) $P(S/B) = 1$

iii) $P(A_1 \cup A_2/B) = P(A_1/B) + P(A_2/B)$ (provided that the events A_1 and A_2 are mutually exclusive).

Let us now apply this concept to a real-world example:

EXAMPLE-2

At a certain elementary school in a Western country, the school-record of the past ten years shows that 75% of the students come from a two-parent home, and that 20% of the students are low-achievers and belong to two-parent homes.

What is the probability that such a randomly selected student will be a low achiever GIVEN THAT he or she comes from a two-parent home?

SOLUTION

Let A denote a low achiever and B a student from a two-parent home.

Applying the relative frequency definition of probability, we have

$$P(B) = 0.75 \text{ and } P(A \cap B) = 0.20.$$

Thus, we obtain

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.20}{0.75} = 0.27$$

And now, let us consider the Multiplication Theorem of probability:

It is interesting to note that the multiplication theorem is obtained very conveniently from the formula of conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

As discussed earlier, the conditional probability of A given that B has occurred has already been defined as:

$$P(A / B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0$$

Multiplying both sides by $P(B)$, we get

$$P(A \cap B) = P(B) \cdot P(A/B).$$

And if we interchange the roles of A and B, we obtain:

$$P(A \cap B) = P(A) P(B/A), \text{ provided } P(A) > 0.$$

MULTIPLICATION LAW

If A and B are any two events defined in a sample space S, then

$$\begin{aligned} P(A \cap B) &= P(A) P(B/A), \text{ provided } P(A) > 0, \\ &= P(B) P(A/B), \text{ provided } P(B) > 0. \end{aligned}$$

(The second form is easily obtained by interchanging A and B.)

This is called the GENERAL rule of multiplication of probabilities.

It can be stated as follows:

“The probability that two events A and B will both occur is equal to the probability that one of the events will occur multiplied by the conditional probability that the other event will occur given that the first event has already occurred.”

$$P(A \cap B) = P(A) P(B/A),$$

Let us apply the concept of multiplication theorem to an example:

A box contains 15 items, 4 of which are defective and 11 is good. Two items are selected. What is the probability that the first is good and the second defective?

Let A represent the event that the first item selected is good, and B, the event that the second items is defective.

Then we need to calculate the probability of the JOINT event $A \cap B$ by the rule $P(A \cap B) = P(A)P(B/A)$.

We have:

Type of Item	No. of Items
Defective	4
Good	11
Total	15

Since all the items are equally likely to be chosen, hence
 $P(A) = 11/15$.

Given the event A has occurred, there remain 14 items of which 4 are defective.

Therefore the probability of selecting a defective item after a good item has been selected is $4/14$ i.e.

$$P(B/A) = 4/14.$$

Hence

$$\begin{aligned} P(A \cap B) &= P(A)P(B/A) \\ &= 11/15 \times 4/14 \\ &= 44/210 \\ &= 0.16. \end{aligned}$$

Example

Suppose a card is drawn at random from a deck of ordinary playing cards.

What is the probability that it is a diamond, a face card or a king?

Solution

Let

A = the card drawn is a diamond

B = the card drawn is a face card, and

C = the card drawn is a king.

Since the events A, B, C are not mutually exclusive, hence the required probability is given by:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) \\ &\quad - P(B \cap C) + P(A \cap B \cap C) \end{aligned}$$

Now, in this example, we have:

$$P(A) = 13/52$$

$$P(B) = 12/52$$

$$P(C) = 4/52$$

Also:

$$\begin{aligned} P(A \cap B) &= P(A) P(B/A) \\ &= 13/52 \times 3/13 = 3/52 \end{aligned}$$

(This is so, because, once a card of diamond has been drawn, the conditional probability of obtaining a face card given that the card is a diamond is $3/13$, the reason being that there are only 3 face cards among the 13 cards of diamonds.)

Similarly:

$$\begin{aligned} P(B \cap C) &= P(B)P(C/B) \\ &= 12/52 \times 4/12 = 4/52 \end{aligned}$$

The conditional probability of obtaining a king given that the card is a face card is $4/12$, the reason being that there are only four kings among the 12 face cards.

And

$$\begin{aligned} P(A \cap C) &= P(A)P(C/A) \\ &= 13/52 \times 1/13 = 1/52 \end{aligned}$$

The conditional probability of obtaining a king given that the card is a diamond is $1/13$, the reason being that there is only 1 king among the 13 diamond cards.

Also

$$P(A \cap B \cap C) = P(A)P(B/A)P(C/A \cap B)$$

This is the extension of the general multiplication theorem to the case of two events, and it states that the probability that A occurs and B occurs, and C occurs multiplied by the probability that B occurs given that A has already occurred multiplied by the probability that C occurs given that both A and B have already occurred.

In this example:

$$\begin{aligned}P(A \cap B \cap C) &= P(A)P(B/A)P(C/A \cap B) \\&= 13/52 \times 3/13 \times 1/3 \\&= 1/52\end{aligned}$$

Hence, substituting all the values, we obtain

$$\begin{aligned}P(A \cup B \cup C) &= 13/52 + 12/52 + 4/52 \\&\quad - 3/52 - 4/52 - 1/52 + 1/52 \\&= 22/52 = 0.42\end{aligned}$$

Hence, the probability that the card that is drawn is a diamond, a face card or a king (or any two of them simultaneously, or all three of them simultaneously) is 42%

Consider all families "with two children (not twins). Assume that all the elements of the sample space $\{BB, BG, GB, GG\}$ are equally likely. (Here, for instance, BG denotes the birth sequence "boy girls"). Let A be the event $\{BB\}$ and B be the event that 'at least one boy'. Calculate $P(A | B)$.

Solution : Here,

$$A = \{ BB \}$$

$$B = \{ BB, BG, GB \}$$

$$A \cap B = \{ BB \}$$

$$\therefore P(A \cap B) = \frac{1}{4}$$

$$P(B) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Assume that a certain school contains equal number of female and male students. 5 % of the male population is football players. Find the probability that a randomly selected student is a football player male.

Solution : Let M = Male

F = Football player

We wish to calculate $P(M \cap F)$. From the given data,

$$P(M) = \frac{1}{2} \quad (\because \text{School contains equal number of male and female students})$$

$$P(F|M) = 0.05$$

But from definition of conditional probability, we have

$$P(F|M) = \frac{P(M \cap F)}{P(M)}$$

$$\begin{aligned} \Rightarrow P(M \cap F) &= P(M) \times P(F|M) \\ &= \frac{1}{2} \times 0.05 = 0.025 \end{aligned}$$

If A and B are two events, such that $P(A) = 0.8$,
 $P(B) = 0.6$, $P(A|B) = 0.5$, find the value of

- (i) $P(A \cup B)$ (ii) $P(B|A)$ (iii) $P(A|B)$.

Solution :(i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= 0.8 + 0.6 - 0.5 = 0.9$$

(ii) $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.5}{0.8} = \frac{5}{8}$

(iii) $P(A | B) = \frac{P(A \cap B)}{P(B)}$

$$= \frac{0.5}{0.6} = \frac{5}{6}$$

Find the chance of drawing 2 white balls in succession from a bag containing 5 red and 7 white balls, the balls drawn not being replaced.

Solution : Let A be the event that ball drawn is white in the first draw. B be the event that ball drawn is white in the second draw.

$$\therefore P(A \cap B) = P(A)P(B | A)$$

Here,

$$P(A) = \frac{7}{12}, P(B | A) = \frac{6}{11}$$

$$\therefore P(A \cap B) = \frac{7}{12} \times \frac{6}{11} = \frac{7}{22}$$

A coin is tossed until a head appears or until it has been tossed three times. Given that head does not occur on the first toss, what is the probability that coin is tossed three times ?

Solution : Here, it is given that head does not occur on the first toss. That is, we may get the head on the second toss or on the third toss or even no head.

Let B be the event, " no heads on first toss".

Then $B = \{TH, TTH, TTT\}$

These events are mutually exclusive.

$$P(B) = P(TH) + P(TTH) + P(TTT) \quad \dots(1)$$

Now $P(TH) = \frac{1}{4}$ ($\because$ This event has the sample space of four outcomes)

and $P(TTH) = P(TTT) = \frac{1}{8}$ ($\because$ This event has the sample space of eight outcomes)

Putting these values in (1), we get

$$P(B) = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

Let A be the event "coin is tossed three times".

Then $A = \{TTH, TTT\}$

$\therefore$ We have to find $P(A|B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Here, $A \cap B = A$

$$\therefore P(A|B) = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$



THE

END